Teen Mental Health
& Social Media

A Data Analysis Case Study

Dataset: 1,200 Teenagers | Ages 13—19

Tools: PostgreSQL • Power BI

Problem Statement

> *How does social media usage impact the development of teenagers, and what can schools and parents do about it?*

Social media has become an integral part of teenage life. Platforms like Instagram and TikTok now compete directly with sleep, exercise, and face-to-face interaction for a teenager's time and attention. While the benefits of connection and self-expression are real, growing concern exists around the mental health consequences of excessive use.

This case study uses a dataset of 1,200 teenagers aged 13—19 to explore whether social media usage is associated with measurable changes in anxiety, stress, academic performance, sleep, and depression. The goal is not only to identify the patterns but to draw conclusions that could inform practical action by parents, schools, and policymakers.

Data Overview

About the Dataset

The dataset contains 1,200 student records with 13 variables covering social media habits, sleep patterns, academic performance, physical activity, social interaction, and mental health indicators. The data was synthetically generated to reflect realistic trends observed in teen mental health research.

Data Dictionary

The following table describes each column in the dataset:

Column	Description	Scale / Values
`age`	Age of the student	13 — 19 years old
`gender`	Gender of the student	Male or Female
`daily_social_media_hours`	Hours spent on social media per day	Numeric (hours)
`platform_usage`	Primary platform used	Instagram, TikTok, or Both
`sleep_hours`	Hours of sleep per night	Numeric (hours)
`screen_time_before_sleep`	Hours of screen time before bed	Numeric (hours)

Column	Description	Scale / Values
`academic_performance`	Academic grade performance	GPA on a 4.0 scale
`physical_activity`	Hours of physical activity per day	Numeric (hours)
`social_interaction_level`	Self-reported social interaction	Low, Medium, or High
`stress_level`	Self-reported stress rating	Scale of 1 — 10
`anxiety_level`	Self-reported anxiety rating	Scale of 1 — 10
`addiction_level`	Self-reported addiction rating	Scale of 1 — 10
`depression_label`	Clinical depression indicator	Binary: 0 = No, 1 = Yes

Summary Statistics

Before diving into the analysis, here is a snapshot of what the average teenager in this dataset looks like:

Average Daily Habits

Metric	Average
Social Media Hours per Day	**4.02 hours**
Sleep Hours per Night	**7.38 hours**
Screen Time Before Bed	**1.72 hours**
Physical Activity	**1.95 hours**
Academic Performance (GPA)	**3.91 / 4.0**

Mental Health Overview

Metric	Average (out of 10)
Stress Level	**2.94 / 10**
Anxiety Level	**2.31 / 10**
Social Media Addiction	**4.40 / 10**
Depression Rate	**17.6%**

> *The average teen spends 4 hours per day on social media and 1.7 hours on screens before bed, while getting 7.4 hours of sleep. Social media addiction scores averaged 4.40 out of 10 — the highest mental health metric recorded — and 17.6% of students showed signs of depression, nearly 1 in 5 teens.*

Analysis & Findings

The following nine questions are each answered with a SQL query, a results table, and a written interpretation of the findings.

Section 1 — Social Media Habits

Q1. How does social media usage vary across age groups?

This query shows the average daily social media hours for each age from 13 to 19.

```
SELECT
    age,
    ROUND(AVG(daily_social_media_hours), 2) AS avg_sm_hours
FROM "teen_mental_health"
GROUP BY age
ORDER BY age;
```

Results:

Age	Avg Social Media Hours
13	2.61
14	3.03
15	3.51
16	4.25
17	4.50
18	4.97
19	5.52

> **Finding:** Social media usage increases consistently with age, rising from 2.61 hours per day at age 13 to 5.52 hours at age 19 — more than double. This steady climb suggests that as teens gain more independence and device access, screen time grows alongside it.

Q2. Does high social media usage correlate with higher anxiety, stress and addiction?

Teens are split into three usage groups and compared across anxiety, stress, and addiction scores (all on a 1–10 scale).

```
SELECT
    CASE
        WHEN daily_social_media_hours >= 5 THEN 'High Usage (5h+)'
        WHEN daily_social_media_hours BETWEEN 3 AND 5 THEN 'Medium Usage (3-5h)'
        ELSE 'Low Usage (<3h)'
    END AS usage_group,
    ROUND(AVG(anxiety_level), 2) AS avg_anxiety,
    ROUND(AVG(stress_level), 2) AS avg_stress,
    ROUND(AVG(addiction_level), 2) AS avg_addiction
FROM "teen_mental_health"
GROUP BY usage_group
ORDER BY avg_anxiety DESC;
```

Results:

Usage Group	Avg Anxiety	Avg Stress	Avg Addiction
High Usage (5h+)	3.13	3.76	5.85
Medium Usage (3-5h)	2.25	2.94	4.34
Low Usage (<3h)	1.55	2.08	3.01

Finding: A clear step-up pattern emerges across all three metrics. High usage teens score double the anxiety of low usage teens (3.13 vs 1.55) and nearly double on addiction (5.85 vs 3.01). This consistent escalation across every usage group strongly suggests more social media time is associated with worse mental health.

Section 2 — Academic & Physical Impact

Q3. Does high social media usage affect academic performance?

Compares average GPA across low, medium, and high social media usage groups.

```
SELECT
    CASE
        WHEN daily_social_media_hours >= 5 THEN 'High Usage (5h+)'
        WHEN daily_social_media_hours BETWEEN 3 AND 5 THEN 'Medium Usage (3-5h)'
        ELSE 'Low Usage (<3h)'
    END AS usage_group,
    ROUND(AVG(academic_performance), 2) AS avg_gpa
FROM "teen_mental_health"
```

```
GROUP BY usage_group
ORDER BY avg_gpa DESC;
```

Results:

Usage Group	Avg GPA
Low Usage (<3h)	3.99
Medium Usage (3-5h)	3.94
High Usage (5h+)	3.79

> **Finding:** Low social media users achieve a GPA of 3.99 compared to 3.79 for high users — a 0.20 point difference that could determine grade boundaries or scholarship eligibility for many students.

Q4. Does screen time before bed affect how much sleep teens get?

Groups teens by bedtime screen habits and compares average sleep hours.

```
SELECT
    CASE
        WHEN screen_time_before_sleep < 1 THEN 'Minimal (<1h)'
        WHEN screen_time_before_sleep BETWEEN 1 AND 2 THEN 'Moderate (1-2h)'
        ELSE 'Heavy (>2h)'
    END AS bedtime_screen,
    ROUND(AVG(sleep_hours), 2) AS avg_sleep
FROM "teen_mental_health"
GROUP BY bedtime_screen
ORDER BY avg_sleep DESC;
```

Results:

Bedtime Screen Time	Avg Sleep Hours
Minimal (<1h)	8.24
Moderate (1-2h)	7.53
Heavy (>2h)	6.75

> **Finding:** Teens with minimal screen time before bed sleep 1.49 more hours per night than heavy screen users (8.24 vs 6.75 hours). Heavy bedtime screen users fall well below the recommended 8–10 hours for teenagers, with downstream effects on mood, concentration, and mental health.

Section 3 — Mental Health Outcomes

Q5. Do depressed teens have a higher connection to social apps?

Compares social media hours, addiction level, and screen time before bed between depressed and non-depressed teens.

```
SELECT
    depression_label,
    ROUND(AVG(daily_social_media_hours), 2) AS avg_sm_hours,
    ROUND(AVG(addiction_level), 2) AS avg_addiction,
    ROUND(AVG(screen_time_before_sleep), 2) AS avg_screen_before_bed
FROM "teen_mental_health"
GROUP BY depression_label
ORDER BY depression_label DESC;
```

Results:

Depression	Avg SM Hours	Avg Addiction	Avg Screen Before Bed
Yes (1)	4.67	4.98	1.98
No (0)	3.88	4.28	1.66

> **Finding:** Every social media metric is elevated in the depressed group — 0.79 more hours per day, 0.70 higher addiction, and 0.32 more hours of screen time before bed. While this does not confirm causation, the consistent pattern across all three metrics is notable.

Q6a. Does social media negatively affect real social interactions? (SM Hours)

```
SELECT
    social_interaction_level,
    ROUND(AVG(daily_social_media_hours), 2) AS avg_sm_hours
FROM "teen_mental_health"
```

```
GROUP BY social_interaction_level
ORDER BY avg_sm_hours DESC;
```

Results:

Social Interaction Level	Avg SM Hours
Low	4.42
Medium	3.97
High	3.77

Q6b. Anxiety and stress by social interaction level

```
SELECT
    social_interaction_level,
    ROUND(AVG(anxiety_level), 2) AS avg_anxiety,
    ROUND(AVG(stress_level), 2) AS avg_stress
FROM "teen_mental_health"
GROUP BY social_interaction_level
ORDER BY avg_anxiety DESC;
```

Results:

Social Interaction Level	Avg Anxiety	Avg Stress
Low	3.02	3.54
Medium	2.22	2.86
High	1.81	2.54

Q6c. Depression rate by social interaction level

```
SELECT
    social_interaction_level,
    ROUND(AVG(depression_label) * 100, 1) AS depression_rate_pct
FROM "teen_mental_health"
GROUP BY social_interaction_level
ORDER BY depression_rate_pct DESC;
```

Results:

Social Interaction Level	Depression Rate %
Low	26.0%
Medium	17.0%
High	9.9%

> **Finding:** Teens with low social interaction have a depression rate of 26% compared to just 9.9% for highly social teens, and anxiety nearly double (3.02 vs 1.81). Social media appears to substitute for real-world interaction rather than replacing its mental health benefits — deepening isolation rather than resolving it.

Q7. Does poor sleep combined with high social media use increase depression risk?

```
SELECT
    CASE
        WHEN daily_social_media_hours >= 5 AND sleep_hours < 6
            THEN 'High SM + Poor Sleep'
        WHEN daily_social_media_hours >= 5 AND sleep_hours >= 6
            THEN 'High SM + OK Sleep'
        WHEN daily_social_media_hours < 5 AND sleep_hours < 6
            THEN 'Low SM + Poor Sleep'
        ELSE 'Low SM + OK Sleep'
    END AS risk_profile,
    ROUND(AVG(depression_label) * 100, 1) AS depression_rate_pct
FROM "teen_mental_health"
GROUP BY risk_profile
ORDER BY depression_rate_pct DESC;
```

Results:

Risk Profile	Depression Rate %
High SM + Poor Sleep	42.9%
High SM + OK Sleep	26.4%
Low SM + Poor Sleep	16.7%
Low SM + OK Sleep	13.4%

Finding: This is the most striking finding in the dataset. Teens combining high social media use with poor sleep have a 42.9% depression rate — more than three times the rate of teens with healthy habits (13.4%). Poor sleep appears to act as a bridge that amplifies the mental health effects of social media.

Section 4 — Protective Factors

Q8a. Does physical activity protect against depression?

```
SELECT
    CASE
        WHEN physical_activity < 1 THEN 'Low (<1h)'
        WHEN physical_activity BETWEEN 1 AND 2 THEN 'Moderate (1-2h)'
        ELSE 'High (>2h)'
    END AS activity_level,
    ROUND(AVG(depression_label) * 100, 1) AS depression_rate_pct
FROM "teen_mental_health"
GROUP BY activity_level
ORDER BY depression_rate_pct DESC;
```

Results:

Activity Level	Depression Rate %
Low (<1h)	29.7%
Moderate (1-2h)	18.9%
High (>2h)	15.1%

Q8b. Anxiety and stress by activity level

```
SELECT
    CASE
        WHEN physical_activity < 1 THEN 'Low (<1h)'
        WHEN physical_activity BETWEEN 1 AND 2 THEN 'Moderate (1-2h)'
        ELSE 'High (>2h)'
    END AS activity_level,
    ROUND(AVG(anxiety_level), 2) AS avg_anxiety,
    ROUND(AVG(stress_level), 2) AS avg_stress
FROM "teen_mental_health"
GROUP BY activity_level
```

```
ORDER BY avg_anxiety DESC;
```

Results:

Activity Level	Avg Anxiety	Avg Stress
Low (<1h)	2.49	3.30
Moderate (1-2h)	2.38	2.99
High (>2h)	2.20	2.86

> **Finding:** Physical activity is the clearest protective factor in the dataset. Teens exercising less than 1 hour per day have a depression rate of 29.7% compared to 15.1% for those exercising more than 2 hours — nearly halving the risk. Anxiety and stress decline steadily as activity increases.

Q9a. Are younger teens more vulnerable to social media addiction?

```
SELECT
    age,
    ROUND(AVG(addiction_level), 2) AS avg_addiction,
    ROUND(AVG(anxiety_level), 2) AS avg_anxiety
FROM "teen_mental_health"
GROUP BY age
ORDER BY age;
```

Results:

Age	Avg Addiction	Avg Anxiety
13	3.5	1.7
14	3.8	1.9
15	4.1	2.0
16	4.6	2.4
17	4.7	2.5
18	5.1	2.8
19	5.3	3.1

Q9b. Depression rate by age

```
SELECT
    age,
    ROUND(AVG(depression_label) * 100, 1) AS depression_rate_pct
FROM "teen_mental_health"
GROUP BY age
ORDER BY age;
```

Results:

Age	Depression Rate %
13	11.0
14	11.0
15	15.0
16	18.4
17	18.2
18	25.6
19	23.5

Finding: Addiction, anxiety and depression all increase steadily with age. Depression rates more than double from 11% at age 13 to 25.6% at age 18. This reflects cumulative exposure to social media growing with age — and so do its effects.

Power BI Dashboard

The following three pages from the Power BI dashboard visualise the key findings of this analysis. Each page tells a distinct part of the story — from overall trends, through the impacts of social media, to what interventions can help.

Page 1 — Social Media and Teen Mental Health Overview

This page establishes the overall picture. The KPI cards at the top show that the average teen spends 4 hours per day on social media, sleeps 7.38 hours, and achieves a GPA of 3.91. The three charts below reveal the most important age-based trend — every single metric gets worse as teens get older. Social media hours climb from 2.61 at age 13 to 5.52 at age 19. Addiction and anxiety levels rise in parallel, and the depression rate nearly triples from 11% to 25.6%.

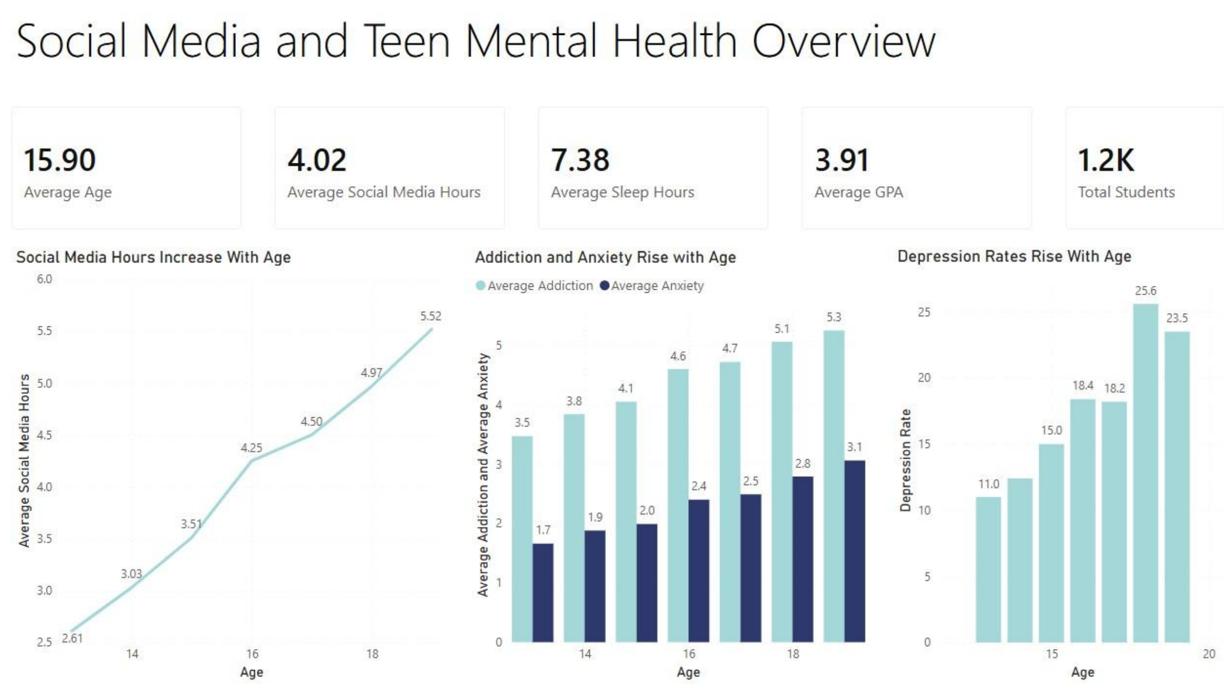

Figure 1 — Social Media and Teen Mental Health Overview

> **Key Finding:** Every metric worsens with age. Social media hours double, depression rates nearly triple, and anxiety climbs steadily — suggesting cumulative exposure to social media compounds its effects over time.

Page 2 — The Impacts of Social Media

This page directly answers the core question of the case study. Three charts examine what happens to mental health, academic performance, and sleep as social media usage increases — and the pattern is consistent across all three. High social media users score 5.9 on addiction vs 3.0 for low users, while anxiety rises from 1.6 to 3.1. GPA declines from 3.99 to 3.79. Most strikingly, teens with heavy screen time before bed sleep just 6.75 hours compared to 8.24 hours for minimal screen users — a difference of 1.49 hours per night.

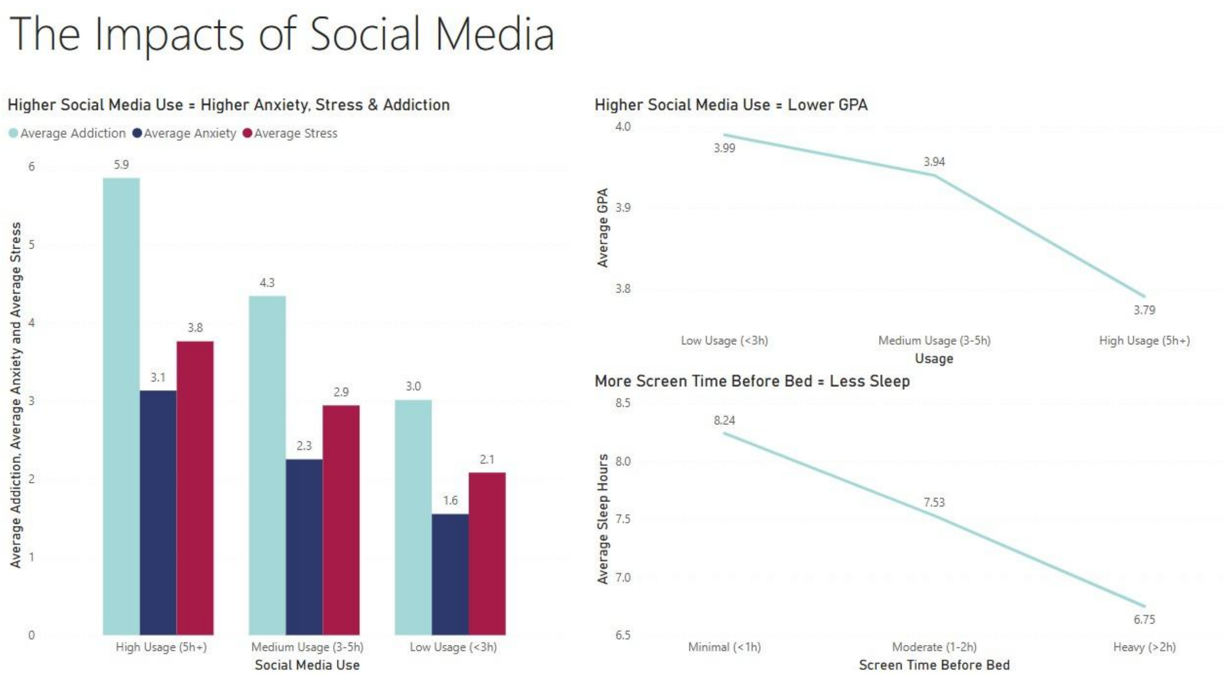

Figure 2 — The Impacts of Social Media

> **Key Finding:** Higher social media use is consistently associated with worse outcomes across every dimension — mental health, academic performance, and sleep — with no exceptions across any usage group.

Page 3 — What Can Help?

This page shifts focus from the problem to solutions. The risk profile chart is the most striking finding in the entire dataset — teens who combine high social media use with poor sleep have a depression rate of 43%, more than three times the rate of teens with healthy habits (13%). The physical activity charts show an equally clear story: teens who exercise less than 1 hour per day have a depression rate of 29.7% compared to 15.1% for those exercising more than 2 hours. Anxiety and stress decline steadily as activity increases.

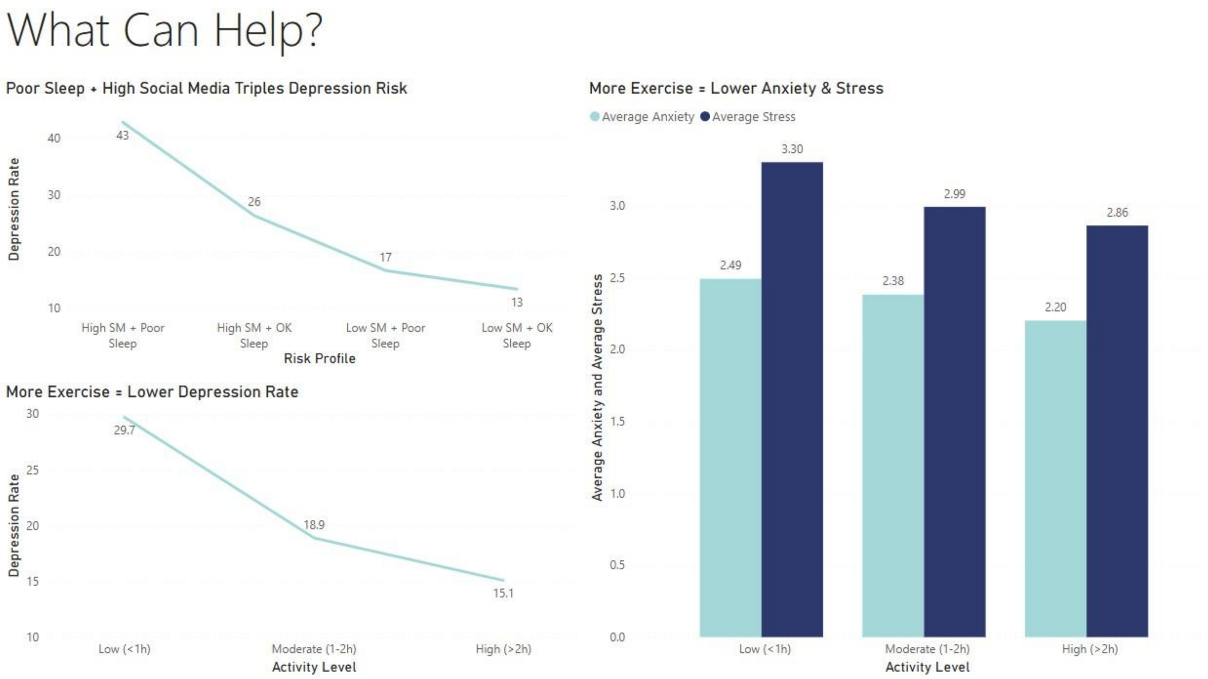

Figure 3 — What Can Help?

Key Finding: Improving sleep and increasing physical activity are the two strongest and most actionable protective factors identified in this analysis. Together with reducing social media use, they form a clear three-part intervention strategy.

Conclusion & Recommendations

Summary of Findings

This analysis of 1,200 teenagers aged 13—19 reveals a consistent and concerning pattern: as social media use increases, so does anxiety, addiction, stress, and depression — while sleep and academic performance decline. These trends worsen steadily with age, suggesting that cumulative exposure to social media over time compounds its effects on teen mental health.

> **Important:** This study demonstrates correlation rather than causation. The data cannot prove that social media directly causes depression or anxiety — only that they consistently appear together. However, the patterns are strong and consistent enough across every metric to warrant serious attention from parents, schools, and policymakers.

Key Findings

- **Every metric worsens with age —** Social media use, addiction, anxiety and depression all increase steadily — depression rates nearly triple from age 13 to 18
- **High SM users have double the anxiety —** Anxiety scores of 3.13 vs 1.55 for low users — a 100% increase
- **Screen time steals sleep —** Heavy bedtime screen users sleep 1.49 fewer hours per night than minimal users
- **Social media hurts grades —** High users average a GPA of 3.79 vs 3.99 for low users
- **Poor sleep amplifies depression risk —** High SM + poor sleep = 42.9% depression rate vs 13.4% for healthy habits
- **Exercise nearly halves depression risk —** From 29.7% for inactive teens to 15.1% for active ones

The Highest Risk Profile

The data identifies a clear high-risk teen who needs the most urgent attention:

Characteristic	Profile
Age	17—19 years old
Daily Social Media Use	5+ hours per day
Sleep	Under 6 hours per night
Physical Activity	Less than 1 hour per day
Depression Rate in This Group	42.9%

These teens are the furthest along in their social media habits and often the hardest to reach through traditional top-down interventions. They require targeted, peer-led approaches that respect their autonomy while creating genuine awareness of the risks they are facing.

Recommendations

The three strongest and most actionable interventions identified by this data are:

Recommendation	Evidence	How to Implement
Reduce Social Media Use	High usage teens have double the anxiety of low users (3.13 vs 1.55)	Phone-free school hours, parental screen time agreements, app usage limits
Improve Sleep Habits	Poor sleep + high SM = 42.9% depression vs 13.4% for healthy habits	No screens 1hr before bed, consistent bedtimes, sleep education in schools
Increase Physical Activity	Active teens have nearly half the depression rate of inactive teens	Mandatory daily PE, after-school sports, making exercise social and fun

Recommendations by Age Group

Younger Teens (13—15) — Structural Interventions

At this age, habits are still forming and external structure is most effective. Teachers and parents have the greatest influence during these years.

- **Daily PE —** Mandatory physical education built into the school schedule directly addresses the exercise gap identified in the data
- **Phone-free school hours —** Restricting phones during school reduces overall daily usage and protects both focus and sleep
- **Digital literacy classes —** Teaching younger teens how social media algorithms are designed to maximise engagement builds early critical awareness
- **Parental screen time agreements —** Setting phone-free times before bed — such as no screens after 9pm — directly targets the sleep finding

Older Teens (16—19) — Autonomy & Logical Reasoning

Older teens respond poorly to being told what to do. Autonomy, peer influence, and data-driven reasoning are more effective at this age.

- **Show them the data —** Presenting findings like 'teens who sleep 8 hours have half the depression rate' is more persuasive to a 17 year old than a rule
- **Peer-led mental health programs —** Older teens are more influenced by peers than authority figures — student-led wellness initiatives are more effective than top-down programs
- **Sleep as a performance tool —** Framing sleep as a performance enhancer for grades and sport — rather than a mental health intervention — resonates better with this age group
- **App usage dashboards —** Encouraging teens to set personal screen time limits respects their autonomy while creating genuine self-awareness
- **After-school sports and clubs —** Making exercise social and competitive rather than mandatory is far more effective for older teens

Closing Statement

> *The most important takeaway from this data is that no single intervention is sufficient. The highest-risk teens are struggling across multiple dimensions simultaneously — high social media use, poor sleep, and low physical activity all reinforce each other in a cycle that is difficult to break alone. The most effective approach addresses all three levers together: reduce social media, improve sleep, and increase exercise. Even small incremental improvements across all three could meaningfully reduce the 42.9% depression rate seen in the highest-risk group — and given that nearly 1 in 5 teens in this dataset show signs of depression, the case for action is clear.*